

such as KVM, AWS, VMware, Eucalyptus Cloud, OpenStack and CloudStack. It also supports deployment of Big Data and analytics environments such as Hadoop, Riak and Aerospike.

Salt

Salt is an open source multitasking CM and remote execution tool. It has a Python-based approach to represent infrastructure as a code philosophy. The remote execution engine is the heart of Salt. It creates a high speed and bi-directional communication network for a group of resources. A Salt state is a fast and flexible CM system on top of the communication system provided by the remote execution engine. It is a CLI-based tool. You can use Git to install it. It provides an asynchronous file server to speed up file serving. It also offers a number of modules to address operating systems, tools/applications and cloud platforms. The user can create custom modules in PyDSL or Python. Halite is a Web UI of Salt. Salt has a highly modular and extensible design to support diverse applications. The Salt system has different module types to manage specific actions such as remote execution and state management behaviour.

The modules can be classified into six groups:

- Execution modules are used for direct execution from the remote execution engine
- State modules are used for the back-end of the Salt CM system
- Grains are used for detecting static information about a system
- Renderers are used to render the information passed to the Salt state system
- Returners allow the return information generated by the remote execution to be returned to an arbitrary location
- Runners are applications executed by the Salt-run command

The dependencies required to run Salt are Python 2.6 >= 2.6 <3.0, msgpack-python, YAML, Jinja2, MarkupSafe, apache-libcloud, Requests – HTTP library, mako and gcc.

Vagrant

Vagrant is an open source CM tool for building and managing easy-to-configure, reproducible and portable virtual development environments. It works on top of virtualisation products such as KVM, VirtualBox, VMware and configuration management tools such as Chef, Puppet, Salt, etc. it is written in Ruby and supports various programming languages such as Java, PHP, C#, Python and JavaScript. It provides support for the libvirt plugin named vagrant-libvirt.

To install Vagrant, download it from <https://www.vagrantup.com/downloads> based on the platform you are using, and install it following the regular procedure prescribed for your system's OS.

Vagrant isolates dependencies and configuration within a single disposable, efficient and consistent environment. It also provides consistent workflow for developing and testing

infrastructure management scripts.

Bcfg2

Bcfg2 or 'bee-config' is an open source fifth-generation CM tool that assists in configuration analysis, visualising configuration information, configuration auditing, service deployment and reporting tools. It has a central configuration model that enables administrators to manage the configuration of compute resources efficiently. Bcfg2 uses the XML model for configuration specification. Thus, Bcfg2 uses XML schema validators and the custom schemas included in Bcfg2 to validate configuration specifications. It is portable and can be executed on GNU/Linux distributions such as openSUSE, Red Hat/RHEL, Fedora, Gentoo, CentOS, Debian, Ubuntu, FreeBSD, OpenBSD, OpenSolaris, Solaris, AIX, Mac OS X, etc. It enables consistent, repeatable and verifiable descriptions of the target environment. Bcfg2 has client-server architecture.

Bcfg2 client:

- It interprets the literal configuration provided by the Bcfg2 server
- It performs all client configuration or reconfiguration
- It does not need client-side processing of the configuration
- It uploads a set of statistics to the Bcfg2 server
- It is aware about how to implement declarative configuration modifications



Figure 6: Bcfg2

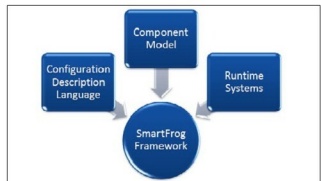


Figure 7: SmartFrog components